

ALIF RIDHWAN AL GHOZALI

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PROFESSIONAL SUMMARY

Detail-oriented Computer Engineering student at Institut Teknologi Sepuluh Nopember (ITS) with a strong foundation in full-stack software development, systems integration, and network infrastructure. Proven ability to design and implement robust enterprise automation tools and complex embedded systems hardware-software logic. Adept at data structure optimization and technical troubleshooting, seeking to contribute high initiative and core technical expertise as a Software Engineer Intern at BFI Finance.

EDUCATION

Institut Teknologi Sepuluh Nopember (ITS)

Surabaya, Indonesia

Bachelor of Engineering in Computer Engineering (Teknik Komputer)

Expected Sep 2026

- Student ID (NRP): 5024221034 | Academic Supervisor: Arta Kusuma Hernanda, S.T., M.T.
- **Relevant Coursework:** Object-Oriented Programming (Java), Database Systems (SQL), Advanced Computer Networks, Algorithm Design & Data Structures, Telematics Prototyping.

PROFESSIONAL EXPERIENCE

PT PLN Nusantara Power UP Brantas

Malang, Indonesia

IT Infrastructure & System Integration Intern (Engineering Division)

Dec 2025 – Feb 2026

- **Web Application Development (Company Profile):** Architected and developed a comprehensive corporate profile web application equipped with an integrated admin panel for seamless data management. Designed a highly responsive UI to dynamically showcase complex enterprise data, including organizational structures, operational statistics (286 MW capacity across 13 PLTA units), performance metrics, and ESG commitments.
- **System Automation & Portal Development (SAKTI):** Engineered SAKTI (Sistem Akun & Tanda Tangan Integritas), an integrated internal self-service portal designed to streamline IT administrative workflows. Developed automated backend logic to handle corporate password reset requests and the digital signing of integrity pacts, effectively digitizing manual processes and enhancing operational efficiency.
- **Enterprise Network Architecture:** Designed and mapped a high-availability, multi-site network topology for the power plant's infrastructure. Visualized the interconnections between the central Core Switch, Fortinet firewalls, and remote ICON+ routers across various operational sites (e.g., PLTA Sutami, Sengguruh, Siman), ensuring secure and robust data routing capabilities.

PT Winnicode Garuda Teknologi

Surabaya, Indonesia

Full-Stack Software Engineer Intern

Mar 2025 – Aug 2025

- **Full-Stack Development:** Engineered the end-to-end architecture of the NEW N WOW NEWS web portal, developing a highly responsive user interface with React.js, Vite, and Tailwind CSS alongside a robust backend server within a unified repository.
- **Backend & Database Architecture:** Developed secure server-side infrastructure using Node.js, Express.js, and JWT for secure user authentication, integrated with a PostgreSQL database to manage relational structures across users, news entries, and interactive comments.
- **System Integration & Logic:** Implemented dynamic client-side routing, state management, and administrative modules (including content creation via AddNews), ensuring seamless data communication between the React frontend and the Node.js backend environment.

TECHNICAL PROJECTS PORTFOLIO

Electromechanical Refreshable Braille Module (Telematics Engineering Project)

ITS Surabaya

Hardware-Software Integration Lead

Sep 2025 – Jul 2026

- Engineered an IoT assistive technology prototype from end-to-end, integrating ESP32 microcontrollers, custom solenoids, and N52 neodymium magnets.
- Designed and modeled precision 3D-printed mechanical housing components using Autodesk Fusion to accommodate micro-actuators with sub-millimeter tolerances.
- Authored C++ firmware to ensure real-time synchronization between digital text inputs and physical solenoid actuation.
- Conducted rigorous hardware stress-testing to calibrate electromagnetic timing, eliminating pin latency and ensuring responsive tactile feedback.

Implementation of Qwen Language Model Based on Retrieval-Augmented Generation (RAG) for Automatic Question Answering Solution in Helpdesk Service (Tugas Akhir / Final Thesis Project)

ITS Surabaya

Backend & AI Software Engineer

Sep 2025 – Jul 2026

- Engineered a Dual-Mode Retrieval-Augmented Generation (RAG) pipeline using Node.js and a quantized local LLM (Qwen3:1.7B via Ollama), optimizing inference allocation for resource-constrained environments.
- Overhauled the offline data ingestion process by implementing Structured JSON Chunking and building an in-memory Flat-File Vector Database (vector_data.json), eliminating external relational database overhead.
- Integrated a dynamic Semantic Automatic Routing gate to intelligently bypass non-academic queries, reducing redundant spatial vector calculations and preserving server memory footprint.
- Slashed document retrieval latency to 44.8 ms (a 60% improvement over conventional chunking) and achieved a 99.52% Retrieval Success Rate utilizing a custom Hybrid Search (Cosine Similarity + Lexical Booster) mechanism.
- Successfully maintained an ultra-low hallucination rate of 2.15% and stable end-to-end processing latency (~1590 ms) entirely within a strictly limited 8 GB RAM local server infrastructure.

ORGANIZATION & LEADERSHIP

Himpunan Mahasiswa Teknik Komputer (HIMATEKKOM) ITS

Surabaya, Indonesia

Head of Internal Affairs Division (DAGRI) – "Titik Transformasi" Cabinet

Feb 2025- Dec 2025

- Coordinated inter-departmental synergy and student synchronization workflows, driving division communication strategies and compiling administrative transition ledger documentations.
- Acted as the primary liaison between the student body and the Computer Engineering department, streamlining bureaucratic procedures and maintaining strategic communication with lecturers and faculty officials.

MAGE (Multimedia and Game Event) X

Surabaya, Indonesia

Head of Subdivision Secure and Licensing

Mar 2024 - Nov 2024

- Led the security and licensing subdivision for a major university-wide technology event, successfully negotiating and securing all necessary legal permits and institutional clearances.
- Designed and enforced comprehensive event security protocols and crowd management strategies, ensuring zero security breaches and a safe environment for all participants.

TECHNICAL SKILLS

Core Programming: JavaScript (ES6+), HTML5, CSS3, Java, C/C++ (Arduino/ESP32 Frameworks)

Frameworks & Databases: ReactJS (Fundamental), SQL Query (CRUD), PostgreSQL, Spring Boot (Concepts)

Tools & Systems: Git, GitHub, Advanced Spreadsheet Scripting, AutoCrat API, Jira/Task Management Tools

Infrastructure/OS: Linux (Zorin OS / Ubuntu Environment), Network Topology Mapping (FTTH, Cisco WLC)